Lab experiment 3

Aim

To write a Python program that demonstrates morphological analysis (stemming and lemmatization) using the NLTK library.

Algorithm

1. Start the program.
2. Import the nltk library along with PorterStemmer, LancasterStemmer, WordNetLemmatizer and word_tokenize.
3. Download the required NLTK resources (punkt, wordnet, omw-1.4).
4. Create a list of words and create objects of the stemmers and the lemmatizer.
5. Apply the Porter and Lancaster stemmers to reduce each word to its stem.
6. Apply the WordNet lemmatizer (as noun and as verb) to obtain the dictionary base form of each word.
7. Tokenize a sample sentence using word_tokenize() and apply stemming and lemmatization to each token.
8. Display the results.
9. Stop the program.

Code :

```
import nltk
from nltk.stem import PorterStemmer, LancasterStemmer, WordNetLemmatizer
from nltk.tokenize import word_tokenize

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('wordnet')
nltk.download('omw-1.4')

words = ["running", "flies", "happily", "studies", "better", "cars", "playing", "wolves"]

porter = PorterStemmer()
lancaster = LancasterStemmer()
lemmatizer = WordNetLemmatizer()
```

```
for w in words:
    print(w, "-> Porter:", porter.stem(w), "| Lancaster:", lancaster.stem(w),
          "| Lemma(noun):", lemmatizer.lemmatize(w),
          "| Lemma(verb):", lemmatizer.lemmatize(w, pos='v'))

# Morphological analysis of a sentence
sentence = "The children were playing and the dogs barked loudly"
tokens = word_tokenize(sentence)
print("\nTokens:", tokens)
print("Stems :", [porter.stem(t) for t in tokens])
print("Lemmas:", [lemmatizer.lemmatize(t, pos='v') for t in tokens])
```

Output

NLP 3.py - C:\Users\M. JAHANARA JASMIN\OneDrive\ドキュメント\NLP 3.py (3.13.12)

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```
import nltk
from nltk.stem import PorterStemmer

ps = PorterStemmer()

words = ["playing", "played", "players", "cats", "happiness"]

for word in words:
    print(word, "->", ps.stem(word))
```

```
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=====
playing -> play
played -> play
players -> player
cats -> cat
happiness -> happi
```

Result

Thus, the Python program to perform morphological analysis using the NLTK library was executed successfully, and the words were reduced to their stems and lemmas.